

IoT-Based Smart Home Automation System Using ESP32

Final Year Design Project Proposal

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(2025)

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Abstract

This project presents an IoT-based Smart Home Automation System built using the ESP32 microcontroller. The system enables users to remotely control home appliances, automate device behaviour using environmental sensors, and monitor device activity through a Flutter mobile application. The solution integrates Firebase authentication for secure access and real-time device synchronization.

The system uses a DHT11 sensor for temperature-based fan automation, a PIR motion sensor for occupancy-based light and fan control, an LDR sensor for curtain automation, and a water-level sensor for automatic pump shutoff. Additional features include timer-based scheduling and a peak-time alert mechanism using a buzzer. A subscription model is introduced: users can view basic statistics for up to two devices, while full analytics require a premium subscription.

The project aims to deliver a cost-effective, scalable, and user-friendly smart home solution that enhances convenience, energy efficiency, and intelligent automation.

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1. Introduction

1.1 Background

Smart home automation has grown rapidly with the advancement of IoT technologies. Traditional systems rely on manual switches or basic timers, lacking intelligent decision-making and centralized control. With microcontrollers like ESP32, cloud platforms such as Firebase, and mobile frameworks like Flutter, it is now possible to build affordable and intelligent automation systems.

This project leverages these technologies to create a unified smart home ecosystem that improves comfort, energy efficiency, and user convenience.

1.2 Introduction to IoT

The Internet of Things (IoT) is a group of interdependent devices (Al-Fuqaha, 2015) that can sense and communicate and perform actions independently. IoT allows the daily appliances, including fans, lights, and TVs, to be supervised and managed with the help of mobile apps or automated logic. The IoT systems comprise, as a rule, sensors, actuators, microcontrollers, communication modules, and cloud services.

Figure 1 shows IoT Layer Architecture.

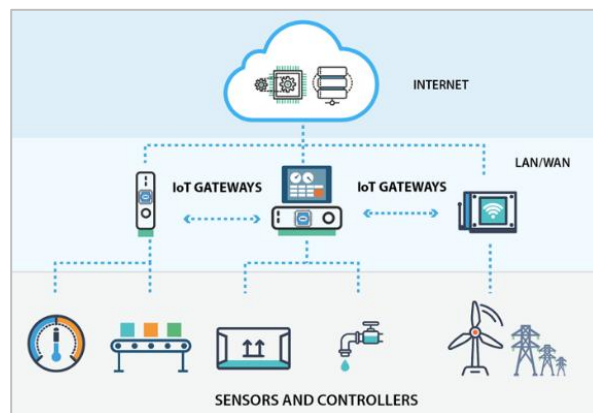


Figure 1: IoT Layer Architecture

1.3 Characteristics of Constrained Devices

IoT devices often operate under constraints (Kaur, 2020) such as:

- Limited processing power
- Restricted memory
- Low energy availability
- Unreliable wireless connectivity
- Small packet sizes and low data rates

These constraints influence protocol design, communication patterns, and system architecture.

1.4 IETF IoT Protocol Stack

The Internet Engineering Task Force (IETF) defines a lightweight protocol stack (Association, 2025) for IoT, including:

- **MQTT**
- **HTTP/REST**
- **WebSockets**
- **TCP/UDP**

In this project, Firebase Realtime Database is used for cloud communication, while ESP32 handles Wi-Fi connectivity.

1.5 Low-Power and Lossy Networks (LLNs)

LLNs consist of devices with limited power, memory, and processing capabilities (Chen, Wang, & Zhang, 2025). They operate over wireless links that may experience interference, packet loss, and limited range. LLNs are widely used in smart homes, sensor networks, and industrial IoT.

1.6 Limitations of LLNs

LLNs face several challenges:

- High packet loss
- Limited bandwidth
- Interference
- Short communication range
- Energy constraints

These limitations must be considered when designing reliable smart home systems.

1.7 ESP32 Overview

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth. It supports multiple sensors, real-time communication, and cloud integration. In this project, ESP32 handles:

- Relay-based appliance control
- Temperature, motion, light, and water-level sensing
- Automatic Mode logic
- Timer scheduling
- Peak-time alerting
- Firebase communication

Figure 2 shows ESP32 structure.



Figure 2: ESP32

1.8 Problem Statement

Households often struggle to manage multiple appliances efficiently. Forgetting to turn off devices, wasting electricity during peak hours, and lacking centralized control are common issues. Existing commercial systems are expensive and require proprietary hardware. There is a need for a smart home system that:

- Controls ESP32-connected appliances remotely
- Automates devices using environmental sensors
- Provides timer-based and peak-time alert functionality
- Offers subscription-based analytics
- Ensures secure access through Firebase
- Provides a unified mobile interface

This project addresses these gaps by developing an integrated, affordable, and intelligent home automation solution.

1.9 Stakeholders & Interests

Understanding who benefits from this smart home system is important because each group interacts with it differently. Homeowners care about convenience and energy savings, renters prefer easy installation, and developers focus on reliability and scalability. Identifying these needs helps shape a system that is practical, usable, and relevant. The main stakeholders are listed in **Table 1**.

Table 1: Stakeholders & Interests

Stakeholder	Interests/Notes	Estimated Market Size
Homeowners	Convenience, automation, energy savings	500,000 - 1,000,000 households
Families	Comfort, ease of use, safety	Included above
Renters	Easy installation, low cost	300,000 – 500,000 households

Developers	IoT + mobile development experience	N/A
Utility Providers	Encourage energy-efficient behavior	N/A

1.10 Objectives

The objectives of the project are as follows:

- Develop a unified ESP32-based smart home system
- Implement secure Firebase authentication
- Enable sensor-driven Automatic Mode
- Provide timer-based device scheduling
- Implement peak-time buzzer alerts
- Introduce subscription-based analytics
- Build a user-friendly Flutter mobile application
- Ensure real-time communication between app, Firebase, and ESP32

1.11 Scope

The scope of the project is as follows:

1.11.1 Inclusion

The IoT Smart Home Automation System focuses on the design, development, and implementation of a smart home solution that improves energy efficiency, convenience, and sustainability. The system includes:

- Manual ON/OFF control of ESP32-connected appliances
- DHT11 temperature-based fan automation
- PIR motion-based fan/light automation
- Water-level sensor for pump auto-cutoff
- LDR-based curtain automation
- Timer mode
- Peak-time buzzer alerts
- Firebase authentication
- Subscription model (2 devices free, full stats premium)
- Flutter mobile app

1.11.2 Exclusion

The project does not cover:

- Voice control
- Wi-Fi smart TV modules
- Alexa/Google Home integration

- AI-based prediction models
- Industrial automation

2. Literature Review

2.1 Related Work

Research highlights the importance of IoT-based automation for energy efficiency and convenience. Research highlights that microcontrollers like the ESP32 can reliably control appliances and communicate with cloud platforms through Wi-Fi (Ahmad, 2023). Sensor-based automation using temperature, motion, and light sensors is widely used to reduce manual effort and support smart decision-making (Nguyen, Patel, & Kumar, 2025).

Commercial systems such as Google Nest and Amazon Alexa offer advanced features but are expensive and often require proprietary hardware, making them less suitable for low-cost implementations (Kaur, 2020).

2.2 Gap Analysis

Existing solutions still have limitations:

- Many low-cost systems lack multi-sensor automation
- Limited support for timer-based control and peak-time alerts
- Commercial systems are costly and not easily customizable
- Few systems offer subscription-based analytics

This project fills these gaps by providing a **low-cost, ESP32-based smart home system** with sensor automation, timer control, peak-time alerts, and a subscription-enabled analytics model.

3. FYDP Overview

This section provides a structured overview of the Final Year Design Project, outlining the project title, team members, and the overall direction of the system. Since the proposed solution integrates Wi-Fi devices, Arduino-controlled appliances, Firebase authentication, Automatic Mode, and voice-based interaction, it is essential to define clear goals, objectives, and success criteria to guide the development process.

3.1 FYDP Title

The project aims to deliver an affordable, scalable, and user-friendly solution tailored for modern residential environments. Its title is as follows:

IoT-Based Smart Home Automation System Using ESP32

3.2 FYDP Architecture

Figure 3 shows the architecture of project we'll adopt.

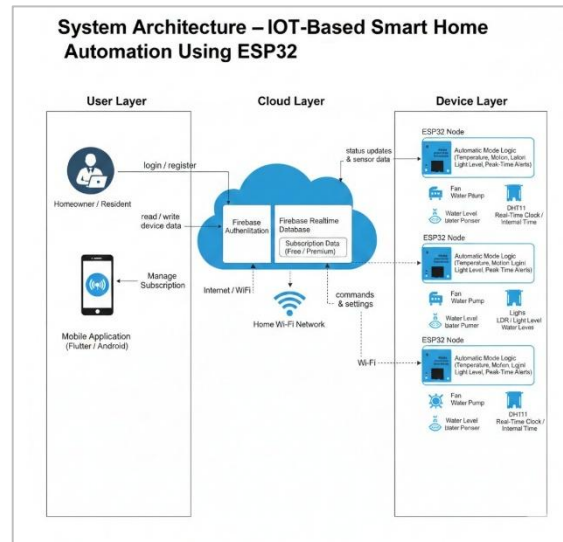


Figure 3: Project Architecture

Our project members are listed below in **Table 2**.

Table 2: Project Members

Sr. No	Roll Numbers	Name	Signatures
1.	BCS22006	Mohsin Saleem	
2.	BCS22016	Abdul Rehman Tahir	
3.	BCS22021	Rehan Ali	
4.	BCS22022	Muhammad Owais	
5.	BCS22024	Wasif Ali	

Table 3 summarizes the goals, assumptions, risks, and target users of the system.

Table 3: FYDP Goals, Objectives, Success Criteria, Assumptions, Risks, and Target Users

FYDP Goals
▪ Develop a unified IoT-based smart home system using ESP32 microcontroller and Wi-Fi communication. ▪ Provide a secure, user-friendly mobile application using Firebase authentication. ▪ Enable intelligent automation through sensors (temperature, motion, light, water level). ▪ Implement timer-based scheduling and peak-time alerting via buzzer. ▪ Introduce a subscription model for device analytics and scalability.

FYDP Objectives
▪ Provide secure user registration and login through Firebase Authentication. ▪ Implement remote ON/OFF control for ESP32-connected appliances. ▪ Introduce Automatic Mode for environment-based decision-making using sensors. ▪ Ensure seamless communication between Flutter app, Firebase, and ESP32 hardware. ▪ Deliver a scalable and cost-effective smart home solution suitable for everyday households. ▪ Integrate subscription logic to differentiate free and premium device analytics.
FYDP Success Criteria
▪ The system reliably controls ESP32-connected devices through the mobile app. ▪ Automatic Mode responds accurately to sensor inputs (temperature, motion, light, water level). ▪ Timer mode executes scheduled operations correctly. ▪ Peak-time buzzer activates during defined hours. ▪ The mobile application is intuitive, stable, and responsive. ▪ Subscription logic functions correctly for device analytics. ▪ Hardware–software integration remains consistent and error-free.
Assumptions
▪ Appliances such as fans, lights, pumps, and curtains can be connected via ESP32 relays. ▪ A stable internet connection is available for Firebase authentication and data sync. ▪ Users follow safety guidelines when interacting with electrical appliances. ▪ ESP32 devices operate reliably in typical home environments.
Risks & Obstacles
▪ Unstable Wi-Fi may affect remote control and Firebase communication. ▪ Hardware failures (sensors, relays, modules) may disrupt automation. ▪ Integration challenges between Flutter, Firebase, and ESP32 may arise. ▪ Incorrect wiring or incompatible appliances may reduce system reliability. ▪ Subscription logic may require additional testing for scalability.
Organization Address: FCIT, University of the Punjab, Lahore, Pakistan
Target End Users
▪ Homeowners seeking convenience, automation, and centralized control. ▪ Families wanting sensor-based smart home features. ▪ Small offices or residential setups needing affordable automation solutions.
Suggested Project Supervisor:
Approved By:
Date:

4. Tools, Libraries and Technologies with Reasoning

This project integrates hardware components, software frameworks, and cloud services to build a unified smart home automation system. The selected tools and technologies ensure

reliable communication between the mobile application, Wi-Fi modules, ESP32 hardware, and Firebase backend. **Table 4** summarizes the tools used in the project along with their rationale.

Table 4: Tools Used

Tools	Tools	Version	Rationale
	ESP32 IDE / Arduino IDE	2.1.1	Used to program ESP32 boards, upload code, and manage sensor/relay control.
	Visual Studio Code (Flutter IDE)	Latest	Lightweight IDE with strong Flutter/Dart support for building the mobile app.
	GitHub	Latest	Version control for collaboration, Code tracking and project management.

Table 5 shows the flutter libraries used in the project along with their rationale.

Table 5: Flutter Libraries

Libraries (Flutter)	Libraries	Version	Rationale
	Flutter Core Libraries	Latest	Handles UI, navigation, and device-control screens.
	Firebase Authentication Library	Latest	Provides secure login and user identity management.
	Firebase Realtime Database / Firestore	Latest	Stores device states and syncs data between app and hardware.
	State Management Library (Provider / GetX)	Latest	Ensures smooth app state handling and real-time updates.

Table 6 shows the ESP32 libraries used in the project along with their rationale.

Table 6: ESP32 Libraries

Libraries (Arduino)	Libraries	Version	Rationale
	ESP32 Wi-Fi Libraries	Latest	Enables Wi-Fi communication for controlling smart devices.
	DHT Sensor Library	Latest	Reads temperature data for Automatic Mode.

	LDR Sensor Library	Latest	Detects light levels for curtain automation.
	PIR Sensor Library	Latest	Detects room occupancy for energy-saving automation.
	Water-Level Sensor Library	Latest	Monitors tank level for pump control.

Table 7 shows the flutter libraries used in the project along with their rationale.

Table 7: Technologies used

Technologies	Technologies	Version	Rationale
	ESP32	N/A	Microcontroller used to control all smart devices and sensors.
	Flutter Framework	3.13	Enables cross-platform mobile app development with responsive UI.
	Breadboard	N/A	Used for prototyping and testing circuit connections.
	Firebase	Latest	Provides authentication and real-time communication between app and ESP32.

5. Work Division

To ensure smooth development and clear responsibility distribution, the project tasks have been divided among team members based on their skills and areas of expertise. The work division is summarized in **Table 8**.

Table 8: Work Division

Sr. No	Roll Number	Name	Role Assignment & Work Division
1.	BCS22006	Muhammad Mohsin Saleem	Mobile App Development: Flutter UI/UX, Firebase integration, Wi-Fi device control screens
2.	BCS22016	Abdul Rehman Tahir	Project Lead / ESP32 development: Sensor integration, relay control, Automatic Mode logic
3.	BCS22021	Rehan Ali	Backend & DevOps: Firebase configuration, database structure, GitHub management

4.	BCS22022	Muhammad Owais	ESP32 Hardware: Circuit Design, sensor wiring, relay modules, ESP32 configuration
5.	BCS22024	Wasif Ali	Mobile app development: Flutter application design, device-control modules, real-time updates, app testing

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